Surprising Privacy Threats from Innocuous Sensors

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ABSTRACT

Smartphones incorporate a plethora of diverse and powerful sensors that enhance user experience. Two such sensors are accelerometer and gyroscope, which measure acceleration in all three spatial dimensions and rotation along the three axes of the smartphone, respectively. These sensors are often used by gaming and fitness apps. Unlike other sensors deemed to carry sensitive user data, such as GPS, camera, and microphone, the accelerometer and gyroscope do not require user permission on Android. This paper presents our IRB-approved studies showing that the these sensors gather sufficient data to quickly infer user's biological sex within a minute of holding the phone. Other approaches to inferring the user's sex rely directly or indirectly on significantly longer time periods. We started with 36 in-person participants, with 89% accuracy, and followed up with 256 on-line participants to show the effectiveness of our technique. We have implemented our technique as transparent background activities in an Android app, and in a Javascript browser web-app for Android and iOS smartphones. We show that even a low-skilled attacker, without expertise in signal processing or deep learning, can succeed at inferring the user's sex. Our approach is device agnostic and does not require tampering with the victim's device or specialized hardware; all study participants used their own phones.

CCS CONCEPTS

• **Security and privacy** → **Mobile and wireless security**; **Privacy protections**; **Usability in security and privacy**; • **Human-centered computing** → *Smartphones.*

KEYWORDS

Privacy, Side channels, Mobile sensors, Demographic Inference

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1 INTRODUCTION

Most mobile devices, including smartphones, tablets, and smartwatches, come equipped with built-in sensors. The accelerometer and gyroscope sensors measure acceleration and rotation forces caused by the movements and vibrations of an object. When the smartphone goes from a standstill to any velocity, the accelerometer is designed to respond to the vibrations associated with such movements. Similarly, when the smartphone rotates about any of the smartphone axes, the gyroscope captures that change.

The privacy implications of inferences from smartphone sensors have previously been reviewed in [10]. For example, [22] showed how to infer the user's gender from their hip movements, gait, and activity patterns while walking for 5-10 minutes with the phone in their pocket. Instead, we observe different patterns in accelerometer and gyroscope data in male and female participants when they hold their phones. We collect accelerometer and gyroscope sensor data from the user's smartphone while the user is reading or browsing the internet, which enables detecting the user's biological sex within sixty seconds with reasonably high accuracy even though the user has not voluntarily specified their sex.

The difference in the holding patterns between the user groups stems from the user's hand stability, attributed to physiologic tremor [13]. Physiologic tremor is an involuntary, rhythmic muscle contraction such as heart beat leading to shaking movements in one or more parts of the body. It occurs in all healthy individuals, is rarely visible to the eye, and typically involves a fine shaking of both of the hands and also the fingers. Tremor is not considered a disease but is a normal human phenomenon that is the result of physical properties in the body [13]. Ergonomics and neurology experts have previously published about physiologic tremor causing hand stability differences due to sex [6, 9], and shown that physiologic tremors can be accurately assessed using only smartphone sensors [1, 4].

Since the hand-stability holding patterns are subliminal in nature, it is difficult to replicate these patterns consciously. The difference is evident in the plots of accelerometer sensor readings from a pair of randomly selected male and female participants from our study, shown in Figure 1. Since a majority of population identify as cisgender [21], for the purpose of communication, we use the terms sex and gender interchangeably in the rest of the paper. The graph shows that the female participant's hand is more stable as compared to that of the male participant.

We conducted experiments with 36 in-person and 256 online smartphone users to demonstrate the effectiveness of our sex-inference technique, showing up to 89% accuracy.

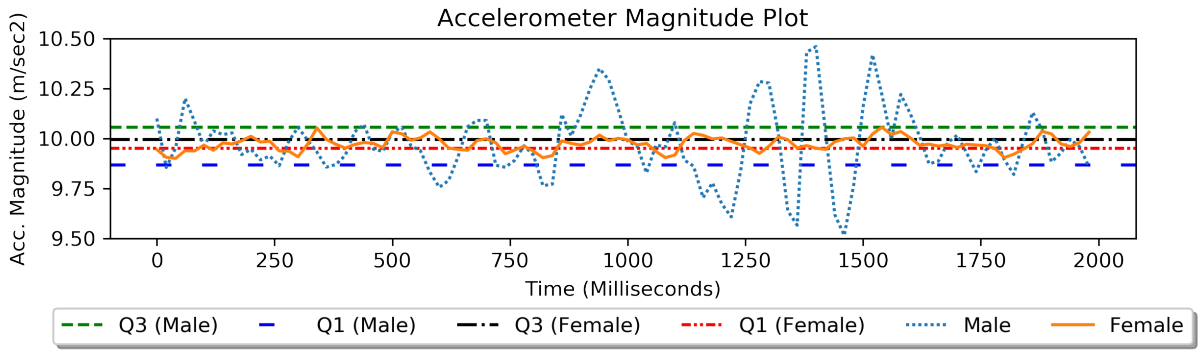

Figure 1: Q1/Q3 = first/third quartile. Accelerometer magnitude timeseries for randomly selected male and female participants shows lower spread in the female's stream, implying less fluctuation while holding the phone.

1.1 Contributions

We evaluate our technique, in terms of both run-time performance and accuracy, and answer these research questions:

- **RQ1**: Can we derive demographic information of the user, specifically the user's sex, from only the accelerometer and gyroscope data streams?
- **RQ2**: Is this a realistic threat? In particular, how much interaction time is needed before making an inference?

The primary contributions of this paper are six-fold:

- We demonstrate that smartphone sensors which, at the time of writing, do not require permission and are available to arbitrary apps, can be used for near-real-time inference of demographic information.
- We demonstrate how an unskilled attacker can leverage off-the-shelf signal-processing & machine learning tools to perform a device-agnostic attack.

To the best of our knowledge, this is the first work on predicting users' demographics in near-real-time (in the order of seconds) that does not require cooperation from the user. In addition, it is the first technique to manipulate physiologic tremor for demographic inference.

The rest of the paper is organized as follows: Section 2 discusses the threat model. Section 3 presents study design and data acquisition methodology. Section 4 provides detailed description of our algorithm for the demographic inference task. Section 5 presents our evaluation scheme and experimental results. Section 6 discusses observations from our study. Section 7 describes limitations and threats to the validity. Section 8 discusses related research efforts to infer demographic information. We conclude with Section 9.

2 THREAT MODEL

We assume that the attacker wants to covertly collect marketable user information without the user knowing this data is being collected, but is still limited by the privacy policies and malware inspection as well as the phone's permission restrictions. The attacker circumvents these controls to obtain the marketable user demographic information by leveraging the data leakage from the built-in motion sensors of a smartphone. Any environment that lets the attacker run code with access to the phone's gyroscope and accelerometer for 30 seconds or longer and allows the attacker to send the collected data across the network functions as an attack vector. Moreover, malware have been shown to use motion and orientation sensors more often than benign apps [11]. In this paper, we show that even a low-skilled attacker, without expertise in signal processing, feature extraction or deep learning, can succeed at gender classification task. Because deep learning algorithms and frameworks have a steep learning curve (many hyper-parameters and network architecture options) and require an advanced skill-set (such as defining the layers, activation functions, regularization, etc.) to train a model, the attacker uses off-the-shelf signal processing libraries and machine learning algorithms to extract relevant features and train models for predictions, respectively. For instance, Random Forest can be used without turning the hyper-parameters. On the other hand, a neural network requires the definition of the network architecture, hidden layers, activation function, regularization, etc. to ensure that it can produce any meaningful results [8, 17]. We demonstrate our proof of concept by collecting data while users read a piece of text, but other possible scenarios include collecting data while users wait for a gaming app to load, or while a user is typing in a messaging app. These heavyweight attack vectors like a maliciously developed app or website present obvious yet realistic threats [23]. The threat model is shown in figure 2. In our proof of concept, we only examine motion and orientation sensor data collected via an Android and web app for demographics inference.

3 STUDY DESIGN

In this section, we describe our process of collecting labeled accelerometer and gyroscope data, which is used for generating and evaluating classification models. Prior to data collection, we obtained approval from our institution's IRB (Institutional Review Board). All subjects provided informed consent before participating in our study and were also asked to fill out a demographics questionnaire. We used IRB-approved deception while collecting the data: Participants were not briefed about the nature of the study or the purpose of data collection to avoid bias. The participants were asked to answer several demographic questions, some of which were used in our study and some of which were not (e.g. height). Some people started the study but did not complete it, and some

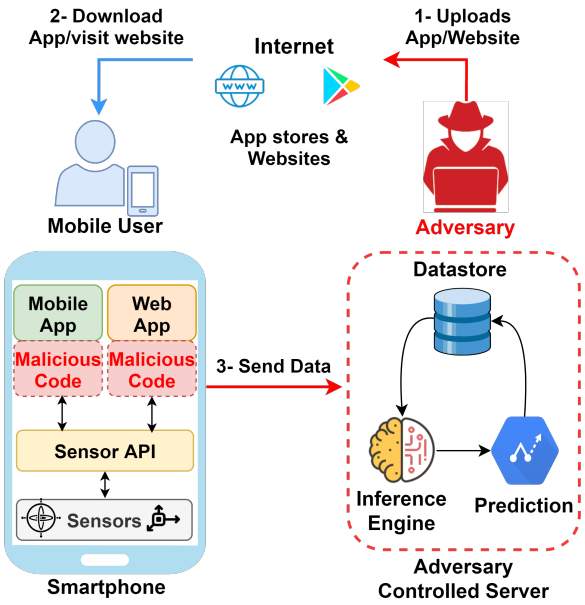

Figure 2: The adversary uploads an app or hosts a website. The user downloads and opens the adversarial app or visits an adversarial website on the browser. The installed app or web-app accesses the accelerometer and gyroscope sensors through API calls and sends the data to the adversary. The adversary can then infer the demographics of the user in real-time. The app implementation could alternatively be self-contained and able to survive intermittent network connectivity, by running the interference engine locally and later sending only the results to the adversary.

did not pass the attention check; these are not included as participants in our data. As a result, we discarded data from about 10% of participants. Ultimately, there were 292 participants for the gender prediction task. The details associated with the collected data set are provided in Section 3.2. We conducted the study in two phases. In the first phase, we recruited 36 participants (17 male and 19 female) for an in-person study. The volunteers consisted entirely of undergraduate and graduate students. In this phase, we collected age, gender, and hand preference of the participants. In the second phase, we recruited 127 on-line participants to collect data using the Android app. We deployed the web app to collect data from an additional 129 on-line users using iPhone and Android devices. In both the phases, participants were asked to use their own devices to complete the study and were given clear instructions on how to perform the task.

3.1 Data Collection Apps

To collect data, we developed an Android app and a web-app. The web-app method of inference requires HTML5 compliant web browser and continuous internet connectivity as the JavaScript transfers the data to the adversarial server and offloads the inference computation to server. The app version is self-contained: it makes inferences in real-time without network access, and can send results to the adversary later, saving significant network bandwidth.

Every participant was asked to read a short passage on their Android phone while holding the phone in either one hand or both hands, as per their preference. We asked the participants to be comfortable in standing posture without using any arm support. Chapanis et al. noted that hand steadiness is more evident in the standing position compared to seated positions [2]. Both apps consist of three tasks; a demographic survey to collect the ground truth, a reading task, and a quiz. The apps collect accelerometer and gyroscope sensor data to capture the hand movements of participants.

3.2 Participant Recruitment

For data collection, we recruited adult participants who speak English either as a first or second language. We curated the data based on the response to the quiz (attention check) and the availability of the accelerometer and gyroscope sensors on participants' phones. We would consider the data only if the participant answered at least two out of three questions correctly. Some participants' smartphones did not have a gyroscope senor; we disregard all such participants from our study. After curating, the results reported in this paper are based on data from 292 participants; 145 male and 147 female Appendix A, Table 2 summarizes the characteristics of our dataset. Participants were spread out geographically, although most stayed in the US or Europe. On average, male and female participants spent 86 and 81 seconds to read the passage, respectively. We advertised on Prolific (www.prolific.co), a well-known online crowd-sourcing service specifically designed for the scientific community [14], helping us to reach its growing number of users.

4 IMPLEMENTATION

In this section, we describe the implementation of the attack. We infer demographic information by leveraging patterns in the sensor readings extracted from the phone. We train a model for *Gender Inference*.

4.1 Feature Extraction

The accelerometer and gyroscope sensor streams are sequences of observations taken sequentially in time. To use a set of time series as input for supervised machine learning algorithms, each time series needs to be mapped into a well-defined feature space. In this section, we present intuition behind the features that were used for our approach. Because our approach relies on physiologic tremor in hands, we extract features from each sensor stream to capture the subtle hand movements. Appendix B provides list of TsFresh features. We group these features into four categories:

- *Distribution features*: These features are used to describe the distribution of the data, e.g., mean, median.
- *Frequency features*: These features capture the frequency of elements to quantify subtle fluctuation, e.g., peaks.
- *Time domain features*: These features rely specifically on temporal knowledge regarding the data points, such as skewness, energy, etc.

- *Combination features*: These features are a combination of features extracted from the raw X, Y, Z signals, such as the sum of X, Y, Z peaks, entropy, etc.

We gather raw X, Y, and Z axes readings from each sensor. In addition to the raw data, we computed the magnitude of accelerometer and gyroscope from the raw sensor data. Our dataset consists of 58 features: 26 features from accelerometer and gyroscope magnitude each, and six combination features that utilize the features extracted from raw X, Y, and Z axes.

Let $X^{m \times n}$ be the multi-modal time series data comprised of n sensor streams obtained over m time instances. We derive statistical features from the time series data over a rolling window W_d of size d with an overlap of ϕ. We use the feature extraction algorithm outlined in Algorithm 1 to derive the features from $X^{m \times n}$ to train our classifiers.

Algorithm 1: Time-Series Data Feature Extraction

Input : $X^{m \times n}$: n sensor streams obtained during m time instances, $t_1,\ t_2\ \ldots\ t_m$
Output: Extracted Features: $X^{r \times \lambda(n)}$: r is instances of processed data over window size W and $\lambda(n)$ is time-series features
Define : W_d (sliding window of size d), ϕ (overlap window)
Function `Feature_Extraction`($X^{m \times n}$):
  Segment m samples into r segments consisting of W_d samples each and ϕ overlap window
  foreach *segment* $s \in r$ **do**
    compute $\lambda(n) = f_1,\ f_2\ \ldots\ f_j$
  return $X^{r \times \lambda(n)}$

5 EVALUATION

As part of our comprehensive evaluation of our classifiers, we used two different evaluation schemes: Stratified 10-fold cross-validation and Leave-One-Out Cross-Validation (LOOCV). Our primary goal is to identify the demographic attributes of one individual, and hence, we emphasize the results of LOOCV. As mentioned in section 2, we use off-the-shelf machine learning algorithms to train our prediction models, to demonstrate that even an attacker without expertise in deep learning algorithms and frameworks can perform the attack. We tried all the classification algorithms in Scikit-Learn [15], including Random Forest (RF), K-Nearest Neighbour (KNN), Support Vector Machines, Naive Bayes, Logistic Regression, and Decision Tree. Because of space limitations we only report the performance of the best tree-based and instance-based learning algorithms, RF and KNN. We ran our scripts on a system with Intel Core i7-6700HQ 4-core, 32GB RAM, running Ubuntu 18.04 and Python 3.7.3. To evaluate the performance of the model, we use four metrics, namely, accuracy, precision, recall, and F1-score.

5.1 Results

In this section, we present the results of models for gender prediction. We studied several other demographic attributes to demonstrate the effectiveness of our work. For space limitations, we only present the results for gender.

In the first phase, we recruited 36 participants (17 male and 19 female) for an in-person study. For this data set, we used LOOCV and achieved an accuracy of 89% and 86% using RF and KNN algorithms, respectively. For the in-person study, RF model achieved precision, recall, and F1 score of 0.93, 0.82, and 0.87, respectively. These results motivated us to perform a large scale study and observe if the same results hold true for larger populations across different demographics.

n = 292	**Predicted** M	**Predicted** F	
Actual M	119 / 127	26 / 18	145
Actual F	37 / 99	110 / 78	147

Table 1: Confusion matrix for gender prediction using LOOCV on 292 participants: RF, KNN

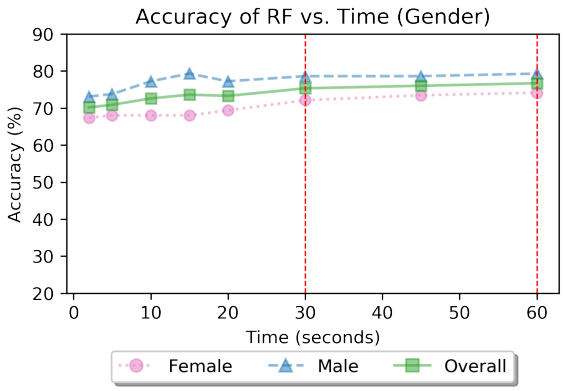

Figure 3: Gender prediction performance over time.

In the second phase, we collected data from 256 online participants. Using LOOCV and RF, we achieved an accuracy, precision, recall, and F1 score of 75.78%, 0.75, 0.78, and 0.76, respectively. Next, we considered data from all 292 participants from in-person study and Prolific. First, we evaluate the performance of RF and KNN algorithms using LOOCV and 10-fold cross validation. We achieved 78.42% overall accuracy on 292 participants using LOOCV. We had recorded an accuracy of 82.06% for male and 74.83% for female participants. RF model achieved precision, recall, and F1 score of 0.76, 0.82, and 0.79, respectively. Table 1 shows the results for both RF and KNN classifiers using LOOCV on all participants. We also applied stratified 10-fold CV to evaluate our technique. The ROC curve of cross-validation is shown in Figure 5. The Area Under the Curve (AUC) is 0.80.

We also studied the minimum user interaction time required to make high confidence inference of gender. Figure 3 shows how the model accuracy increases over time. Time presented in x-axis represents the time spent by the participants on the app. The figure demonstrates how the model's accuracy varies depending on the time spent by the participants on the app. Our model predicted the claimed gender within 15 seconds of user holding the phone with an accuracy of 74%, within 30 seconds with an accuracy of 75%, and within 60 seconds with an accuracy of 77%. We note that these results are lower than for our in-person study. We suspect that

some on-line participants might not have followed the instructions correctly. We also applied t-SNE [12] dimentionality reduction to reduce the feature space to two dimensions. To plot the graph, we used the points utilized by RF algorithm to classify the participants. Figure 4 shows distinction between genders.

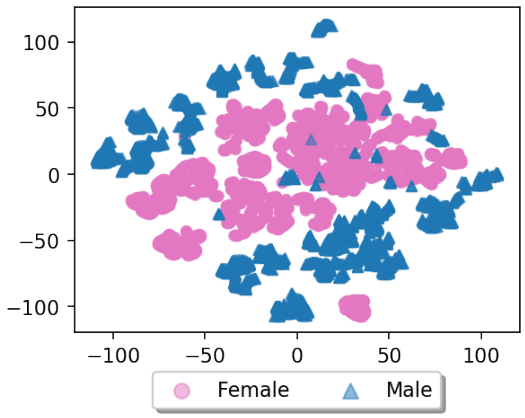

Figure 4: TSNE visualization shows distinction between male and female data points.

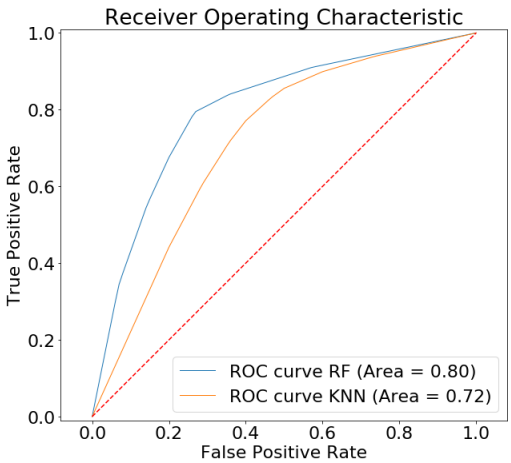

Figure 5: ROC curve of 10-fold CV using RF and KNN

6 DISCUSSION

RQ1: Can we derive demographic attributes of the user by using only accelerometer and gyroscope data?

We demonstrate that using our approach, it is possible to infer demographic information of the smartphone user in real-time by leveraging differences in physiologic tremor. Through supervised learning, our technique can achieve prediction accuracy of up to 78% about the user's gender.

RQ1: We observe that it is possible to infer demographic attributes from the accelerometer and gyroscope readings. In particular, we can predict the gender of the user with high accuracy.

In this paper we demonstrated that the gender prediction task can be performed with high accuracy of 78%. Other potential characteristics that influence physiologic tremors include smoking habits, stress level, alcohol consumption, and substance consumption [13]. We intend to study these features in the future.

RQ2: Is this a realistic threat? In particular, how much interaction time is needed before making an inference?

Our simulations with different phone holding time suggest that the accuracy of gender prediction task increases over time. We achieved optimal results within one minute of the phone holding. We observe that the performance of the model increases over time and the model achieves 76% accuracy within a minute. We show that our approach has significantly lower overhead in terms of time and intrusion level required to make an inference.

RQ2: Gender can be inferred within as little as 5 seconds.

7 LIMITATIONS AND THREATS

In this study, we demonstrate how to predict demographic characteristics of a smartphone user by mining accelerometer and gyroscope sensor data in a time-efficient manner. Our approach assumes that the user is using their smartphone in a position that does not require them to take arm support. We observed that when the participant takes arm support, the pattern in the sensor data diminishes, leading to a higher number of mis-classifications. Moreover, our classifier does not consider the impact of food items containing caffeine. Researchers have observed that participants had higher variations in the hand movement when they regularly consume caffeine (e.g., coffee) [13] or nicotine [5]. We also did not consider the health conditions of the participants or other environmental factors like temperature, which might also influence the results. We leave that to future research.

8 RELATED WORK

Researchers have studied the privacy implications of the accelerometer sensor [10]. Prior studies have been conducted to infer the behavior [19]health parameters, and demographics [16, 20] of the smartphone user. Most of the existing approaches for sensor-based demographic inference in phones are based on gait analysis [3], installed apps [18], browser history [16], and call logs [7]. Other approaches include the processing of visual or audio signals.

The above-mentioned methods of demographic inference require system permissions and historic data from other applications for predictions. Each of these methods is intrusive since they access the user's private information of browser history, call logs, and applications installed on their device. Furthermore, these techniques require substantial data from the device over a long period of time. Hence, they are computationally ineffective. Moreover, above mentioned sensor based techniques require either sophisticated data collection mechanism and/or the data to be collected over long intervals of time. Hence, making them impractical.

9 CONCLUSION

In this paper, we demonstrated that the lack of traditional smartphone privacy mechanisms for the accelerometer and gyroscope sensors leave malicious actors the opportunity to gather private demographic information about the user. We present a non-invasive

transparent technique that infers the user's sex from the data streams of these seemingly innocuous sensors. We conducted an IRB-approved study on 292 smartphone users to demonstrate the effectiveness of our work. Our experimental results show that we can achieve a prediction accuracy of 78% on the user's claimed sex across the total study population of both online and in-person participants, and prediction accuracy of 89% for the user's apparent sex considering only in-person participants. Furthermore, we demonstrate that the interaction time needed to perform such an inference attack with high confidence is within one minute. We provide evidence that even a low-skilled attacker, using only off-the-shelf signal processing and ML libraries, can perform the attack that could potentially be used for data brokerage and political advertising that have far-reaching political and societal impacts.

REFERENCES

[1] Benoit Carignan, Jean-François Daneault, and Christian Duval. Measuring Tremor with a Smartphone. In *Mobile Health Technologies*, volume 1256 of *Methods in Molecular Biology*, pages 359–374. Humana Press, New York NY, December 2014.

[2] Alphonse Chapanis, Wendell R Garner, and Clifford T Morgan. Applied experimental psychology: Human factors in engineering design. 1949.

[3] Sang-Hyun Cho, Jeongsook M. Park, and Oh Yeol Kwon. Gender differences in three dimensional gait analysis data from 98 healthy Korean adults. *Clinical Biomechanics*, 19(2):145–152, February 2004.

[4] Jean-Francois Daneault, Benoit Carignan, Carl Éric Codère, Abbas F Sadikot, and Christian Duval. Using a Smart Phone as a Standalone Platform for Detection and Monitoring of Pathological Tremors. *Frontiers in Human Neuroscience*, 6:357, January 2013.

[5] Dag G Ellingsen, Rita Bast-Pettersen, Jon Efskind, Merete Gjølstad, Raymond Olsen, Yngvar Thomassen, and Paal Molander. Hand tremor related to smoking habits and the consumption of caffeine in male industrial workers. *NeuroToxicology*, 27(4):525–533, July 2006.

[6] Hiroshi Endo and Koichi Kawahara. Relationship between hand stability and the 10-hz physiological tremor during various manual tasks. *Ergonomics*, 53(4):491, April 2010.

[7] Bjarke Felbo, Pål Sundsøy, Alex 'Sandy' Pentland, Sune Lehmann, and Yves-Alexandre de Montjoye. Modeling the Temporal Nature of Human Behavior for Demographics Prediction. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases (ECML PKDD)*, volume 10536 of *Lecture Notes in Computer Science*. Springer, September 2017.

[8] Brett Grossfeld. Deep learning vs machine learning: a simple way to understand the difference | zendesk blog. https://tinyurl.com/yaayltj6, January 2020. (Accessed on 03/14/2020).

[9] Gerald A. Hudgens, Linda T. Fatkin, Patricia A. Billingsley, and Joseph Mazurczak. Hand Steadiness: Effects of Sex, Menstrual Phase, Oral Contraceptives, Practice, and Handgun Weight. *Human Factors*, 30(1):51–60, 1988. PMID: 3350527.

[10] Jacob Leon Kröger, Philip Raschke, and Towhidur Rahman Bhuiyan. Privacy Implications of Accelerometer Data: A Review of Possible Inferences. In *3rd International Conference on Cryptography, Security and Privacy (ICCSP)*, pages 81–87. ACM, January 2019.

[11] Boyu Liu, Duanyue Yun, Xin Guo, Xiao Ji, Huiyu Song, Shirish Singh, and Gail Kaiser. Detecting sensor-based repackaged malware. In *2020 IEEE International Conference on Big Data (Big Data)*, pages 5759–5761, 2020.

[12] Laurens van der Maaten and Geoffrey Hinton. Visualizing Data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, November 2008.

[13] NIH National Institute of Neurological Disorders and Stroke. Tremor Fact Sheet. https://tinyurl.com/y9qtmtdx, March 17 2020. (Accessed on 05/01/2020).

[14] Stefan Palan and Christian Schitter. Prolific.ac—A subject pool for online experiments. *Journal of Behavioral and Experimental Finance*, 17:22–27, March 2018.

[15] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[16] Do Viet Phuong and Tu Minh Phuong. Gender Prediction Using Browsing History. In Van Nam Huynh, Thierry Denoeux, Dang Hung Tran, Anh Cuong Le, and Son Bao Pham, editors, *Knowledge and Systems Engineering*, volume 244 of *Advances in Intelligent Systems and Computing*, pages 271–283. Springer, 2014.

[17] Piotr Plonski. Random Forests® vs Neural Networks: Which is Better, and When? | KDnuggets.com. https://tinyurl.com/ycdskucd, June 2019. (Accessed on 03/14/2020).

[18] Suranga Seneviratne, Aruna Seneviratne, Prasant Mohapatra, and Anirban Mahanti. Your Installed Apps Reveal Your Gender and More! *ACM SIGMOBILE Mobile Computing and Communications Review*, 18(3):55–61, January 2015.

[19] Laura Silver, Aaron Smith, Courtney Johnson, Kyle Taylor, Jingjing Jiang, Monica Anderson, and Lee Rainie. Mobile Connectivity in Emerging Economies | Pew Research Center. https://tinyurl.com/y9o2efxw, March 2019. (Accessed on 06/09/2020).

[20] Shirish Singh, Devu Manikantan Shila, and Gail Kaiser. Side channel attack on smartphone sensors to infer gender of the user: Poster abstract. In *Proceedings of the 17th Conference on Embedded Networked Sensor Systems*, SenSys '19, page 436–437, New York, NY, USA, 2019. Association for Computing Machinery.

[21] Katy Steinmetz. What cisgender means. https://time.com/3636430/cisgender-definition/, 12 2014. (Accessed on 06/02/2021).

[22] Gary M Weiss and Jeffrey W Lockhart. Identifying User Traits by Mining Smart Phone Accelerometer Data. In *5th International Workshop on Knowledge Discovery from Sensor Data (SensorKDD)*, pages 61–69, August 2011.

[23] Jiexin Zhang, Alastair R. Beresford, and Ian Sheret. SensorID: Sensor Calibration Fingerprinting for Smartphones. In *40th IEEE Symposium on Security and Privacy (S&P)*. IEEE, May 2019.

A SUMMARY OF PARTICIPANT DEMOGRAPHICS

Table 2 summarizes the characteristics of our dataset. Participants were spread out geographically, although most stayed in the United States or Europe. On average, male and female participants spent 86 and 81 seconds to read the passage, respectively.

Parameter	Results
Age	Mean: 34 years
	Standard Deviation: 11
	Minimum: 18 years
	Maximum: 76 years
Gender	Male: 145 (49.65%)
	Female: 147 (50.34%)
Preferred Hand	Right Hand: 157
	Left-Hand: 80
	Both-Hands: 55

Table 2: Summary of demographics of study participants. We had 292 participants who participated in two separate data collection rounds.

B TIME-SERIES FEATURES

Following are the Tsfresh features used to compile the training and testing dataset:

- **Distribution features**: Minimum, Maximum, Mean, Median, Standard deviation (SD), Variance, First Quartile (Q1), Third Quartile (Q3), Inter Quartile Range (IQR), Variance larger than standard deviation, Percentage of reoccurring values to all values (PoR), Coefficient of variation
- **Frequency features**: Number of peaks, Count above mean, Count below mean
- **Time domain features**: Skewness, Kurtosis, Energy, Longest strike above mean, Longest strike below mean, Mean abs change (MAC), Mean change, Mean second derivative central, Approximate entropy, Sample entropy, Crest Factor
- **Combination features**: Sum peaks, Sum Sample entropy, Sum Approximate entropy